First observe then perform - children's changing gestural comprehension between the ages of 2 and 5

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1. Research background

In the first years of life, children's vocabulary grows rapidly (Ganger & Brent 2004). This vocabulary is oftentimes accompanied with gestures: both in play and in everyday communication (Clark 2016, Barret 2017). The trajectory of the development of speech and gesture is not linear; children adopt the trial-and-error method in figuring out how signs work (Benson 2020)---sometimes such an approach works well, and allows them to gain new knowledge, sometimes it leads to frustration. Communicative advances can be related to different stages of cognitive and motor development: from speech-oriented to movement-oriented phases (Namy et al. 2004). Research in gesture studies targeting child development looks into whether children understand iconicity (e.g. Tolar et al. 2008), but not into whether there is a difference in how children process iconic gestures of different kinds. In our research, we check whether *enacting* gestures, dynamic ones, resembling an action done with/to an abject, and *representing* gestures, static ones, resembling the shape or structure of an object, are differently comprehended by children at the age of 2, 2.5, 3, 4 and 5. Based on research in child development and imitation skills in children (e.g. Zlatev, 2007, 2013), we assume that *enacting* gestures, related to movement, will score more correct answers compared to *representing* gestures, related to physical characteristics, in all groups. The selection of items was based on McArthur-Bates CDI as well as a pilot with a group of children aged 2.5-4 from a local kindergarten.

1.2. Data collection

We looked at five groups of children (N=50, 10 per group). Each child was presented with a 36-page-long book that contained 4 images per page. Only one image was the target in each page. In 3 rounds, divided by short breaks to keep the children involved, 12 pages per round, each participant was asked to match the gesture performed by the experimenter with an image from the book (Image 1). Each child had to provide 12 answers in *enacting*, 12 answers in *representing*, and 12 answers in speech (which was a control) condition. The task was challenging, because the children were shown two types of iconic

gestures: enacting and representing ones for each object. The research got all of the required consents.

Image 1. The task: left-side picture presenting the gesture performed by the experimenter (top: rowing; bottom: boat-like handshape); right-side picture presenting a child selecting an image.

2. Assumptions

Our aim was to take a more careful look at gestures in child development by extending the study conducted by Tolar et al. (2008) and by breaking down iconic gestures into further sub-types. Across the age groups we analyse, we hypothesised 1) that enacting gestures will be easier to match withn corresponding object, and 2) that iconic gesture comprehension, in spite of its sub-type, stabilises with age and experience.

3. Results

Statistical analyses proved that children obtain better results with age. There also seems to be a change in preference for gesture comprehension: from representing gestures at the age of 24 months to enacting gestures at the age of 36 months; and to stabilisation---ability to understand the meaning of iconic gesture in spite of its type---at the age of 48- and 60-months, when the difference in comprehension is scarcely visible.

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